



USE CASE MODELLING

Film management system

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1. Description of the System Boundary

The proposed system boundary for Cultured Africa determines what the software will be responsible for internally within the company and externally outside the company.

1.1 Internal System Boundary

Refers to the internal components that the web-based system will have, built, managed, and controlled by the team.

Segment	Description
Customer Management System	Registers the user in the system (ensuring they are above the age of 13). Logs users in and out. Handles profile settings.
Administrative Dashboard	Internal portal allowing the Administrator to upload high-quality videos/films. Manages metadata. Views analytics.
Content Database	Structurally stores film descriptions. Stores user data and comments.
Website Streaming Portal	Displays high-quality films and soundtrack playlists.
Payment Integration Layer	Communicates with the payment gateway, verifying that the fee was paid successfully.
Feedback & Moderation	Stores and displays customer comments. Monitors and regulates customer comments.
Search Engine	Interprets user queries to locate specific films.

1.2 External System Boundary

Refers to the external entities and services that will interact with the website application but are not part of the application itself.

Segment	Description
Actors	Users that will be interacting with the website application. The administrator responsible for overseeing the analytics and managing the content.
Third-Party Services	Payment gateways that process transactions. Email or SMS services for verification.
Web Operating Systems	Android (Google) and iOS (Apple) which provide the application a platform to run.

2. Overview of Actors & Goals

The table below provides a detailed overview of the actors that will be interacting with the system and a brief description of what each actor's goals are with the system.

Actor	Main Goals
End-User (Primary Actor)	Access the website to view film content. Register an account and, upon verification, be able to watch the desired film. Provide feedback on film content in the form of written comments and ratings.
Administrator (Primary Actor)	Upload new films and update existing films. Edit film titles and descriptions. Monitor user feedback, thereby fostering a respectful community. Review viewer analytics.
Payment System (Supporting Actor)	Securely process and authorise user payment to unlock a film.

3. Use Case Modelling

3.1 Core Use Cases

Register Account	
Actor	Description
End User	
Main Success Scenario	• A user arrives at the platform home page and clicks "Sign Up" or "Register". • The system displays a registration form. The user enters full name, email address, and password with confirmation. • The system validates input format and checks that the email is not already registered. • The system creates a new user account and sends a verification email. • The system displays a success message. The user is automatically logged in or clicks the verification link. • The system redirects the user to the content browsing page.
Alternative Scenarios	• If the email is already registered, the system displays "An account with this email already exists. Please log in or reset your password" and the use case ends. • If the password does not meet requirements, the system displays password requirements and the user must re-enter a valid password. • If the password confirmation does not match, the system displays an error and the user must re-enter matching passwords. • If a required field is left blank, the system highlights the missing fields and the user must complete all required fields. • If the user cancels registration, the system returns to the home page and no account is created. • If the verification email fails to send, the system logs the error and allows the user to request a new verification email.

Purchase Content	
Actor	Description
End User	

Main Success Scenario	• A registered and logged-in user navigates to a desired film, sees the price and a "Purchase" button. • The user clicks "Purchase" and the system redirects the user to the payment gateway. • The user enters payment information and the payment gateway validates and processes the transaction. • The payment gateway returns success confirmation. The system adds the content to the user's purchased library. • The system displays a success message, triggers a purchase confirmation email, and redirects the user to the content playback page.
Alternative Scenarios	• If the user attempts to purchase already-owned content, the system displays "You already own this content" and offers direct playback. • If the payment is declined, the system displays a decline message, and the user may retry or cancel. • If the payment gateway times out, the system displays an error, no charge is applied, and the user is prompted to retry. • If the content is free, the system bypasses the payment flow and adds the content to the library immediately.

Upload Content	
Actor	Description
	Content Administrator
Main Success Scenario	• An administrator with administrative privileges accesses the admin console and selects the upload function. • The administrator provides the video or audio file and enters metadata (title, description, culture category, and pricing). • The system validates the file type and size, then processes the upload and performs encoding for streaming. • The system confirms successful upload. The content becomes immediately available for browsing and purchase by end users.
Alternative Scenarios	• If the file type is not supported, the system displays an error listing supported formats (such as MP4). • If the file size exceeds the 9GB limit, the system displays an error and the administrator must compress or split the file. • If the upload is interrupted due to network issues, the system pauses the upload and offers a resume option. • If required metadata fields are missing, the system highlights the missing fields and prevents upload until all required information is provided. • If the encoding process fails, the system notifies the administrator and retains the original file for retry.

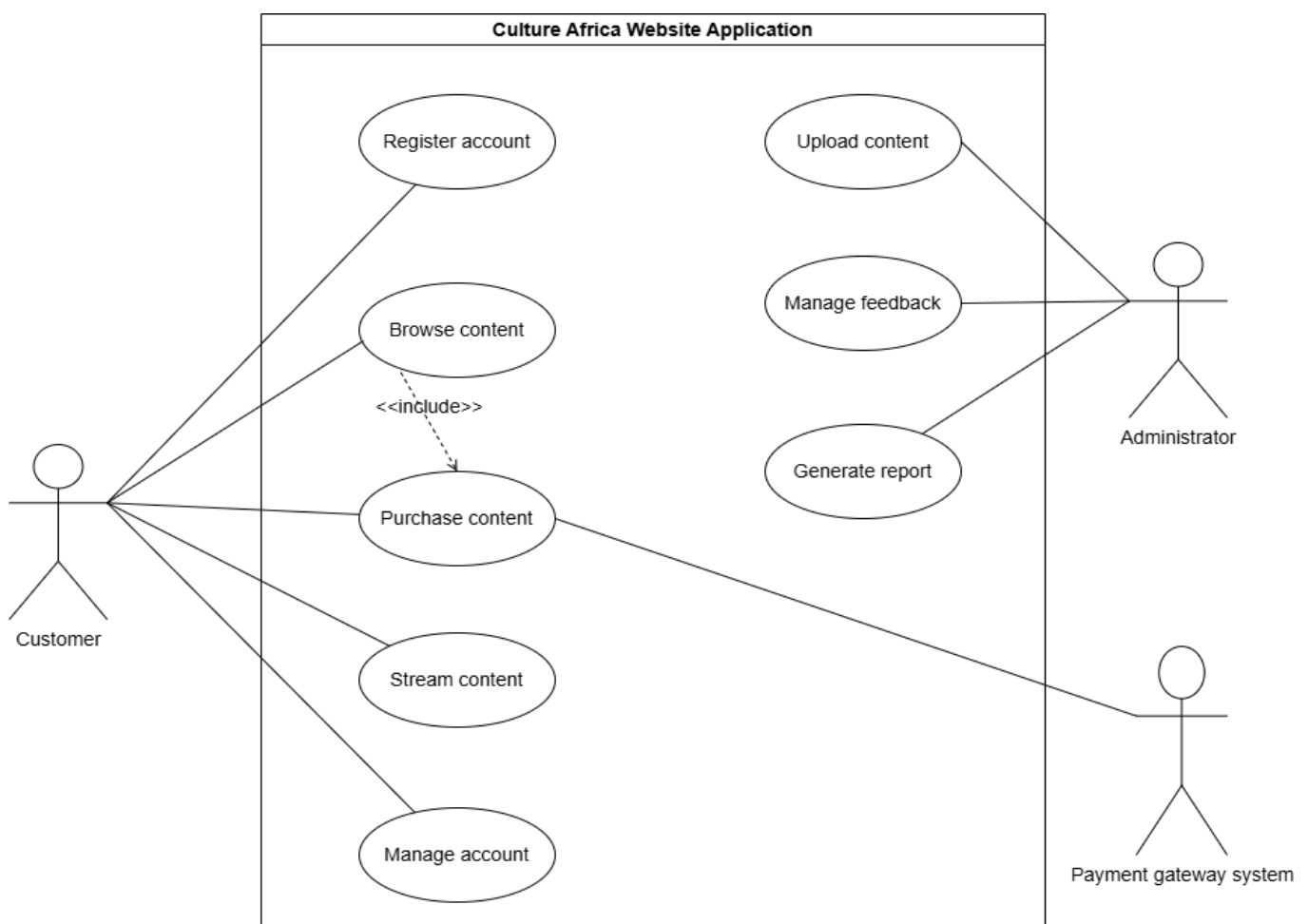
Use Case and Actor	Description
Browse content (End-user)	The end-user navigates to the website's home page to discover available items. Upon arrival, the system presents the user with content that is organized into specific culture categories. The user explores the content by scrolling through thumbnails or applying filters based on criteria like culture, type or price. Selecting a specific item allows the user to view comprehensive details, including descriptions, pricing, and cultural information. Additionally, the system supports keyword searches and sorting options, and it provides feedback if no content matches the chosen filter
Generate report (Administrator)	Content Administrator accesses the system's analytics to create specific reports. The workflow begins with the administrator logging in and navigating to the system dashboard to create a report. After selecting the option to create a report, the administrator chooses the desired report type and specifies the specific details to be included. The system then prompts for a report format, generates the document, and displays the final report to the administrator.
Upload feedback (End-user)	The end-user navigates their previously watched film history to share their thoughts. To initiate the process, the user selects the film and clicks the designated comment button. The user then provides a written comment containing their feedback within the platform. Once the input is complete, the user clicks the post button to upload (post) their comment.

3.2 Maintenance Use Cases

Maintenance Use Case	Description
Manage User Accounts	Performed by the Content Administrator. The administrator can view a list of registered users (join date, purchase count, account status), suspend or activate accounts, and reset user passwords. The user can log in to an existing account or permanently delete the account and associated data for POPIA compliance.
Manage Cultural Categories	Performed by the Content Administrator. The administrator manages the classification of South African cultures represented on the platform — adding new culture categories, viewing all active categories, modifying culture names or descriptions, and removing a culture category.
Manage Content Metadata	Performed by the Content Administrator. The administrator updates metadata for existing content without re-uploading the file — viewing current metadata, modifying title, description, culture category, or pricing, and removing content from the active library by archiving or marking it unavailable.

Maintenance Use Case	Description
Manage Feedback	Performed by the Administrator. The administrator logs into the application and engages with feedback provided by responding to comments, addressing positive comments, and deleting inappropriate comments that are harmful or violate community standards, to encourage a healthy community and dialogue.

3.3 Use Case Diagram



The use case diagram for the Culture Africa Website Application illustrates the interactions between two primary actors: the Customer and the Administrator.

On the user-facing side, the Customer is involved in five core activities:

Register account: The initial process of creating a profile on the platform.

Browse content: Exploring the various cultural items available, as discussed in our previous conversations.

Purchase content: Transacting to gain access to specific items.

Stream content: Viewing or playing the purchased cultural material directly through the application.

Manage account: Handling their personal profile and settings.

On the management side, the Administrator is responsible for three key system functions:

Upload content: Adding new cultural material to the platform for customers to discover.

Manage feedback: Overseeing user comments and interactions, which aligns with the feedback protocols we previously touched upon.

Generate report: Accessing system data to create analytics documents, a process we have also summarized in our discussion

3.3 Use Case Descriptions

Use Case Name	Register Account
Actor	Customer (Primary)
Related Use Cases	N/A
Pre-conditions	The website system is on and functional. The customer doesn't have a registered account.
Post-conditions	The customer successfully registers an account, and account details are saved on the website's cloud.
Flow of activities:	
Basic flow	
Customer:	System:
1.The customer navigates the websites homepage	
2. The customer clicks the "Register" button.	2.1. The system displays the registration form 2.2. The system prompts the customer to enter an email address, create a password and confirm the password.
3. The customer enters their email address, creates a password and confirms the password.	3.1. The system validates the input (required fields and correct data format). 3.2. The system checks if the email address is not already registered. 3.3. The system creates a new user account and sends a verification email.
4. The customer clicks the "Verify email" button to verify their email	
5. The customer is redirected to the website.	5.1. The system prompts the user to enter their email and password
6.The user enters their email and password	6.1The system displays a success message 6.2 The user is successfully logged in to their account.
Alternate flow	
2.1.a. If the customer cancels registration, the system returns to the home page and no account is created.	
3.1.a. If a required field is left blank, the system highlights the missing fields, and the customer must complete all required fields.	
3.1.b. If the password entered does not meet the requirements, the system displays, "password requirements not met: 8 letters and 2 special characters" and the customer must re-enter a valid password.	
3.1.c. If the password confirmation does not match, the system displays an error message "Incorrect password entered" and the customer must re-enter matching passwords.	
3.2.a. If the email is already registered, the system displays "An account with this email already exists. Please log in or reset your password".	
3.3.a. If the verification email fails to send, the system logs the error and allows the customer to request a new verification email.	

Use Case Name	Purchase Content
Actor	Customer (Primary) Payment gateway system (Supporting)
Related Use Cases	Browse content (Includes)
Pre-conditions	The customer is registered and logged in to the website. The content is available on the website.
Post-conditions	The customer successfully purchases the desired film. The film is added to the customer library.
Flow of activities:	
Basic flow	
Customer:	System:
1.The customer navigates the website's home page. 2. The customer selects the desired film. 3. The customer clicks the purchase button. 4. The customer enters payment information (bank card details).	1.1. The system displays content details and price. 3.1. The system redirects the user to the payment gateway. 4.1. The payment gateway validates bank card details. 4.2. The payment gateway processes the transaction. 4.3. The payment gateway returns success confirmation. 4.4. The system adds the content to the customer's purchased library 4.5. The system displays a success message, triggers a purchase confirmation email 4.6. The system redirects the customer to the content playback page.
Alternate flow	
3.1.a. If the customer attempts to purchase already-owned content, the system displays "You already own this content" and offers direct playback. 3.1.b. If the content is free the system bypasses the payment flow and adds the content to the library immediately. 3.1.c If the customer cancels the purchase, the system returns the customer to the homepage. 4.1.a. If the customer provides invalid card details, the payment gateway returns an error message, "transaction failed" 4.1.b. If the payment is declined, the system displays a decline message, and the customer may retry or cancel. 4.2.a. If the payment gateway times out, the system displays an error, no charge is applied and the customer is prompted to retry.	

Use Case Name	Upload Content
Actor	Administrator (Primary) Cloud storage (Support)
Related Use Cases	N/A

Pre-conditions	The administrator gains access to the Administrators dashboard, using appropriate credentials. The administrator is logged in to the system.
Post-conditions	The Administrator successfully uploads content, and it is stored in a cloud and is available in the platform.
Flow of activities:	
Basic flow	
Administrator:	System:
1. The administrator navigates the administrator's dashboard. 2. The administrator provides the video file and enters metadata (title, description, culture category, and pricing)	1.1. The system displays uploading options 1.2. The system prompts the administrator to upload content. 2.1. The system validates the file type and size, then processes the upload 2.2. The system performs encoding for streaming. 2.3. The system confirms successful upload 2.4. The system stores the content in the database. 2.5. The content becomes immediately available for browsing.
Alternate flow	
2.1.a. If required metadata fields are missing, the system highlights the missing fields and prevents upload until all required information is provided. 2.1.b. If the file type is not supported, the system displays an error listing supported formats (such as MP4). 2.1.c. If the encoding process fails, the system notifies the administrator and retains the original file for retry. 2.1.d. If the upload is interrupted, the system pauses the upload and offers a resume option.	

Use Case Reasoning

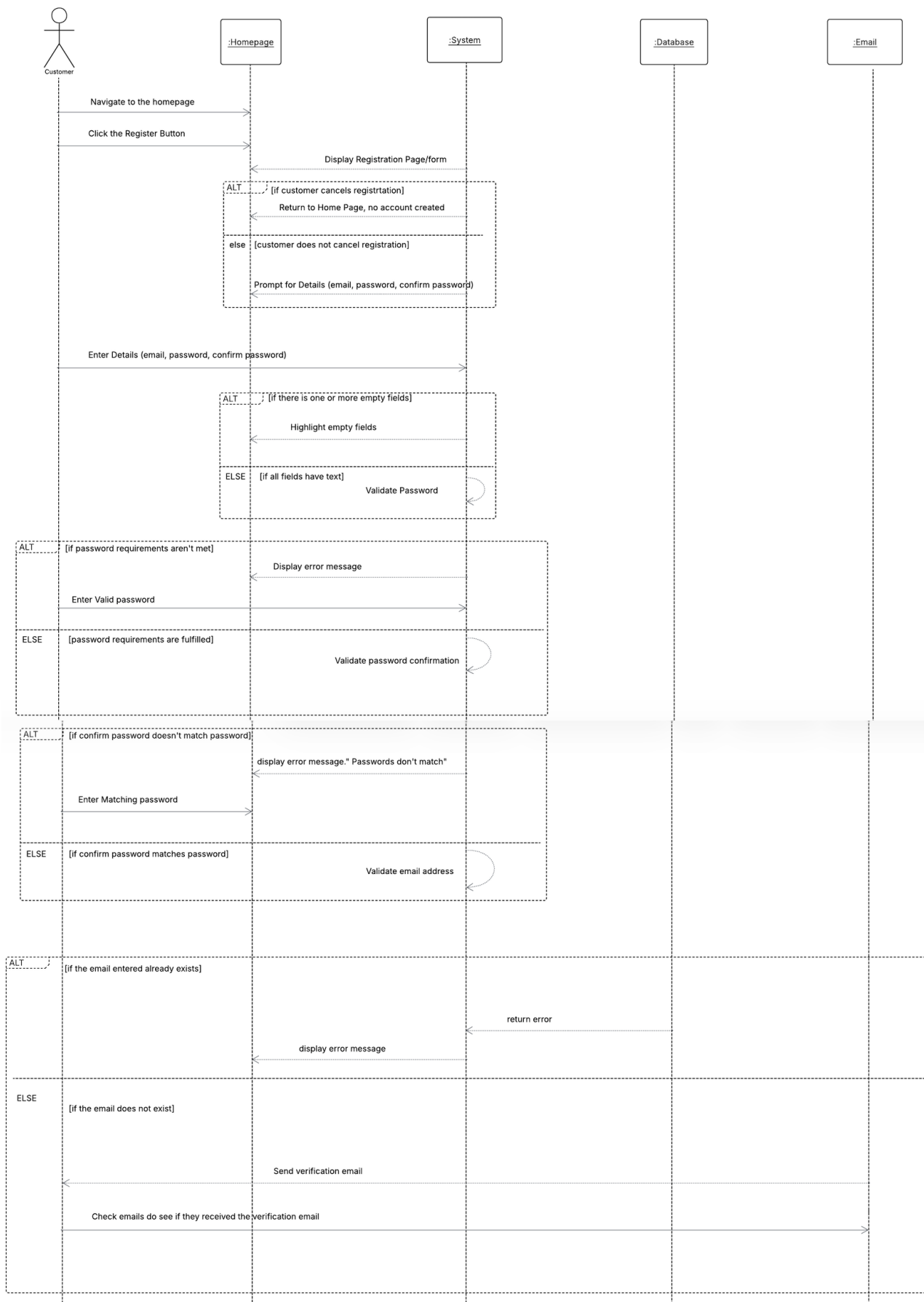
The **Register Account** use case was chosen because it is the entry point into the system. It allows users to create accounts, access personalised features, and securely interact with the platform. Without registration, users would not be able to purchase or access protected content.

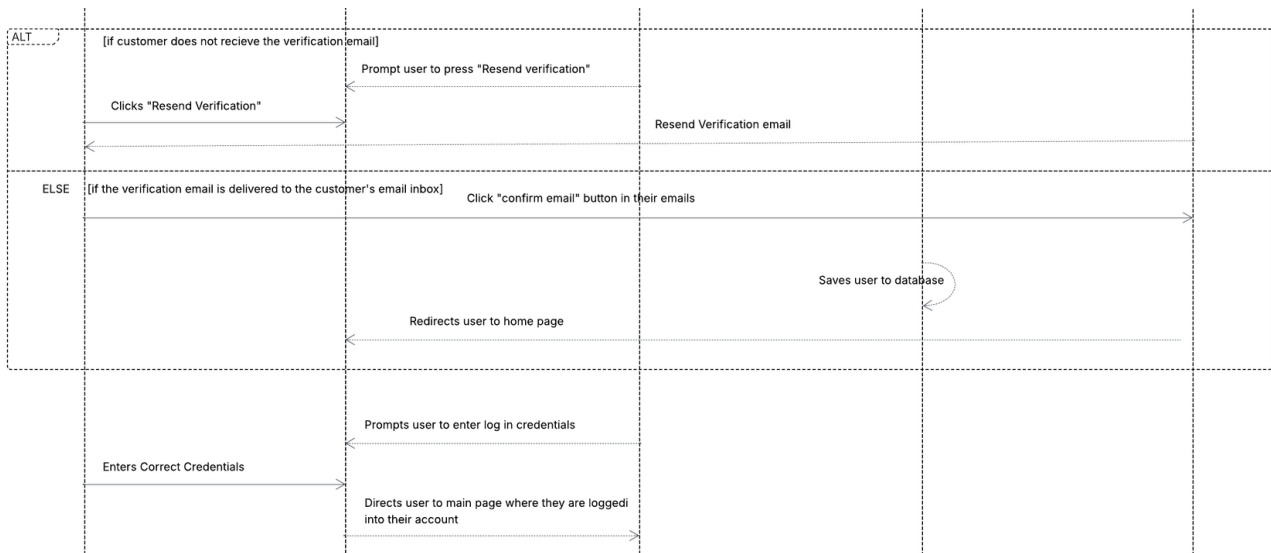
The **Purchase Content** use case was selected because it directly supports the business goal of monetisation. It enables users to unlock films through the wallet-based payment system, which helps Cultured Africa generate revenue from its cultural content.

Together, these two use cases demonstrate both the user access functionality and the revenue-generating capability of the platform, making them strong choices for implementation in this iteration.

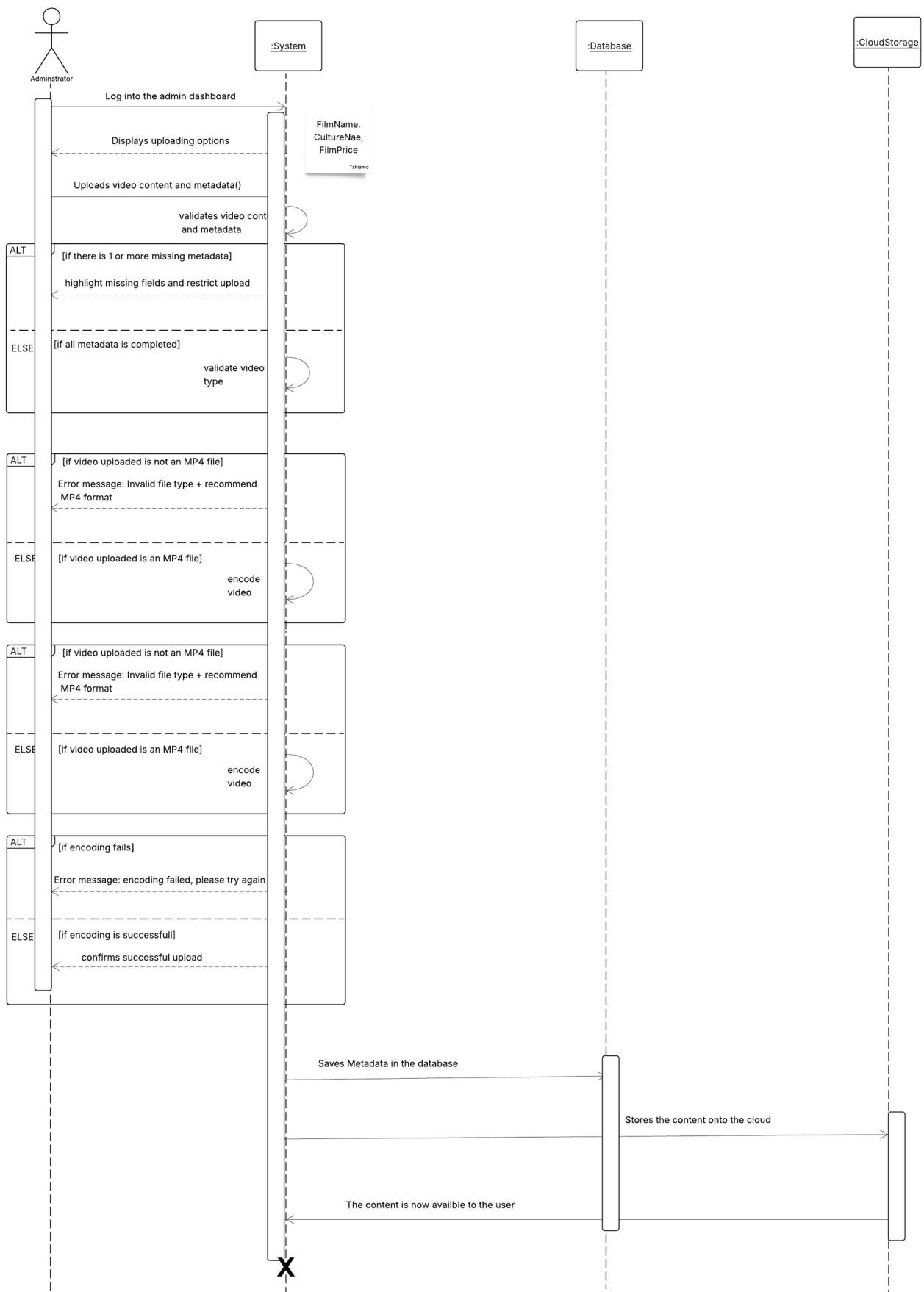
4. Sequence Diagrams

Sequence Diagram: Register Account

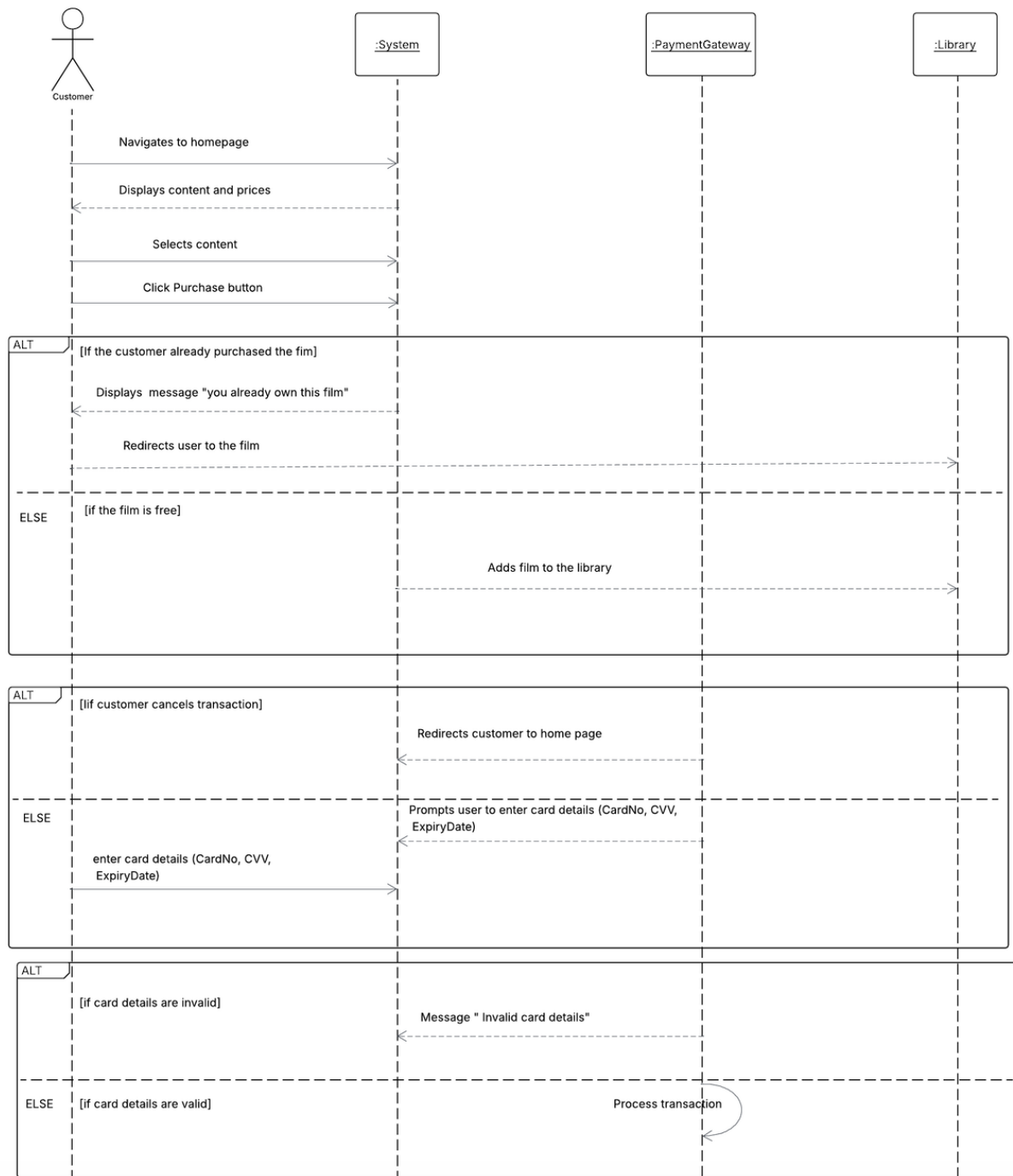


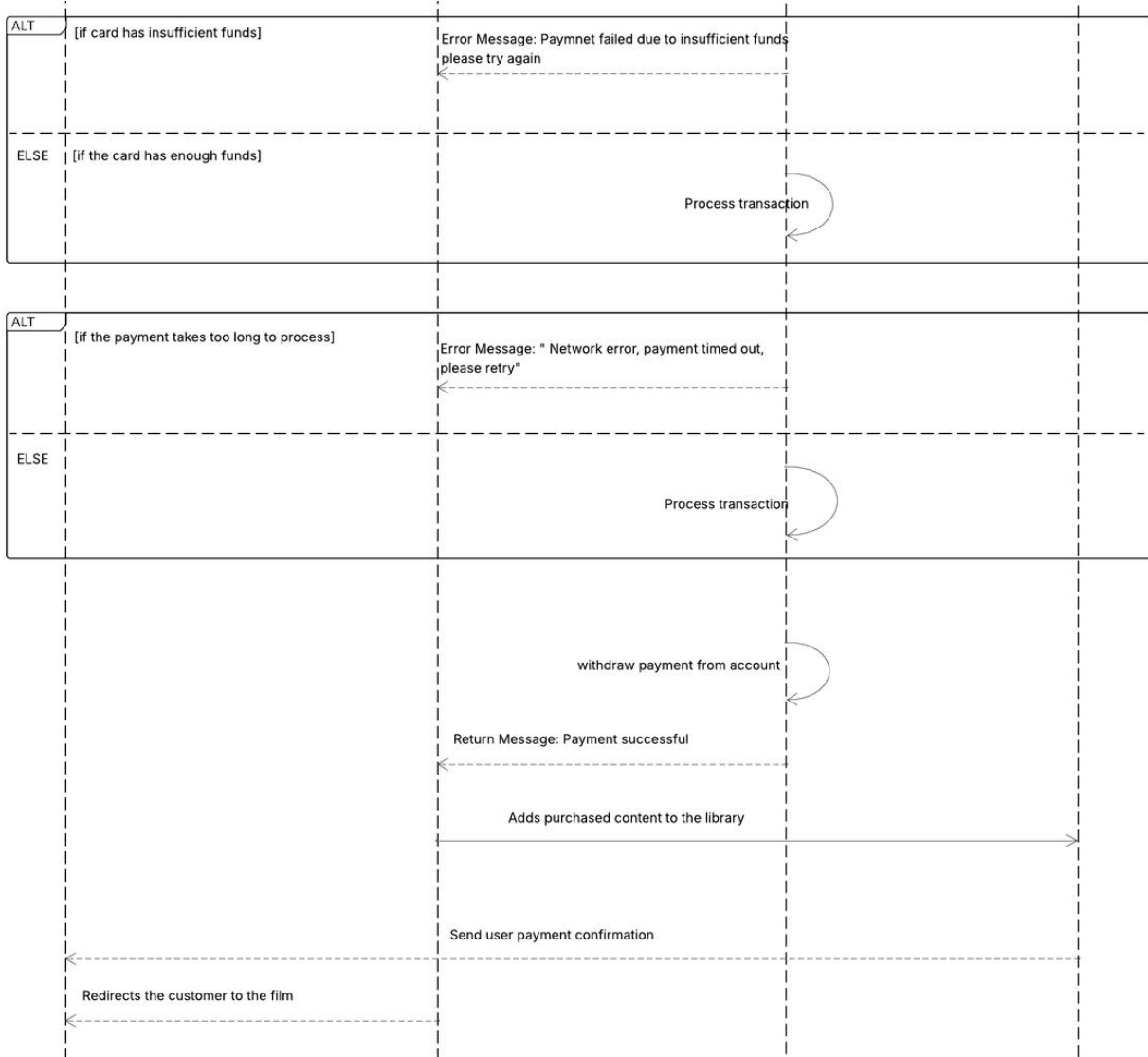


Sequence diagram: Upload Content



Sequence diagram: Purchase Content

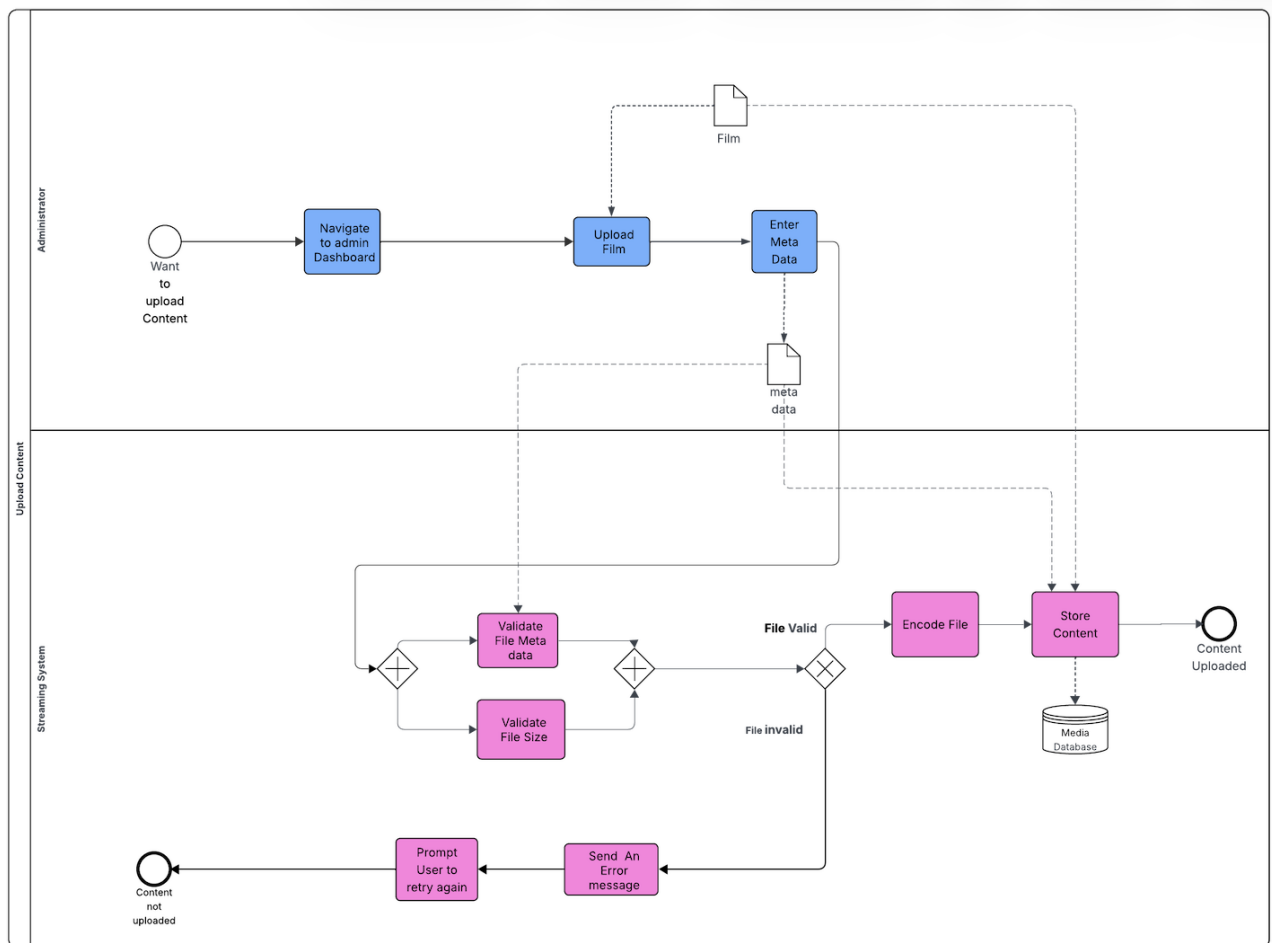




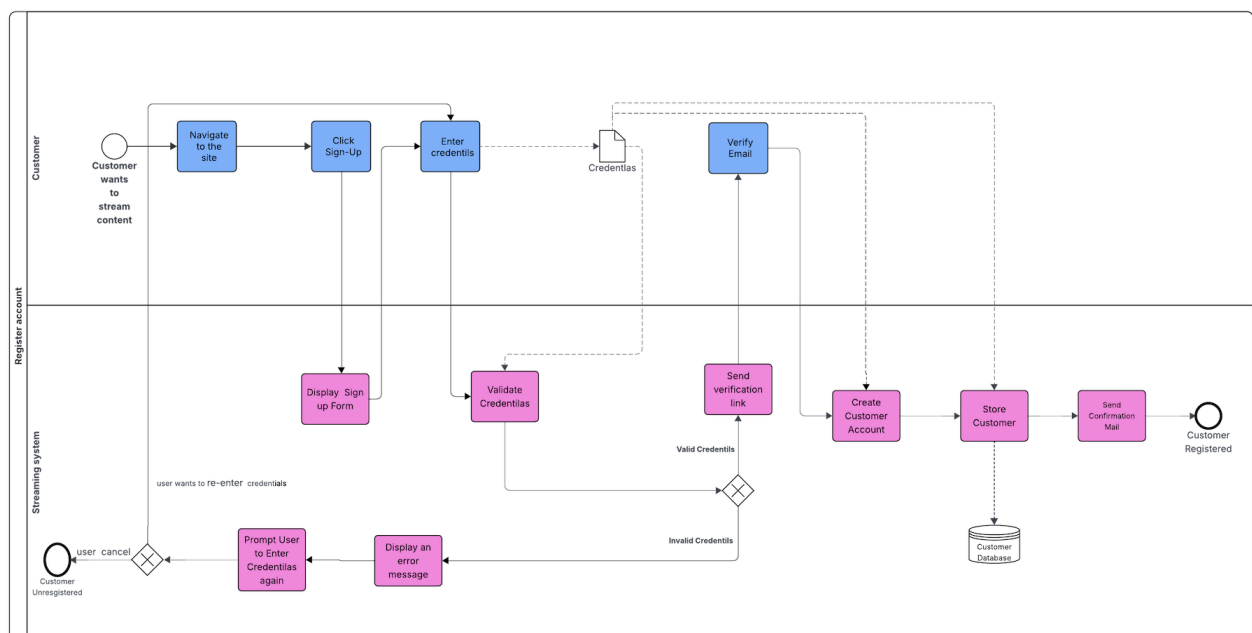
5.BPM Diagrams and Activity Diagram

BPMN Diagram

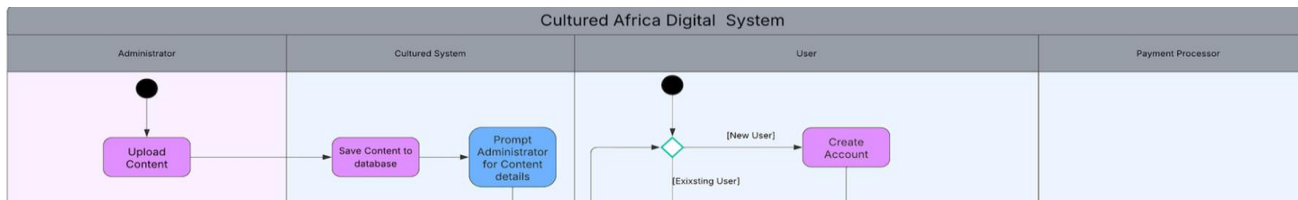
Upload Content Process



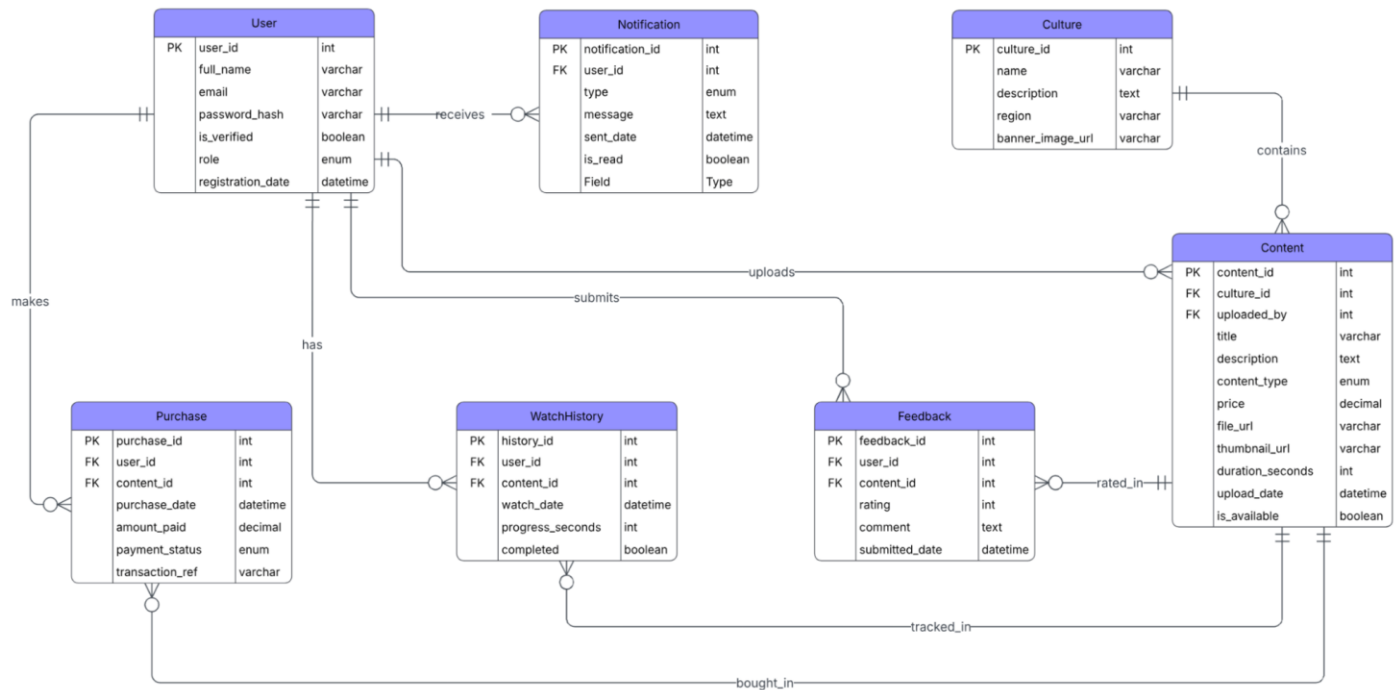
Register Account Process



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Entity Relationship Diagram

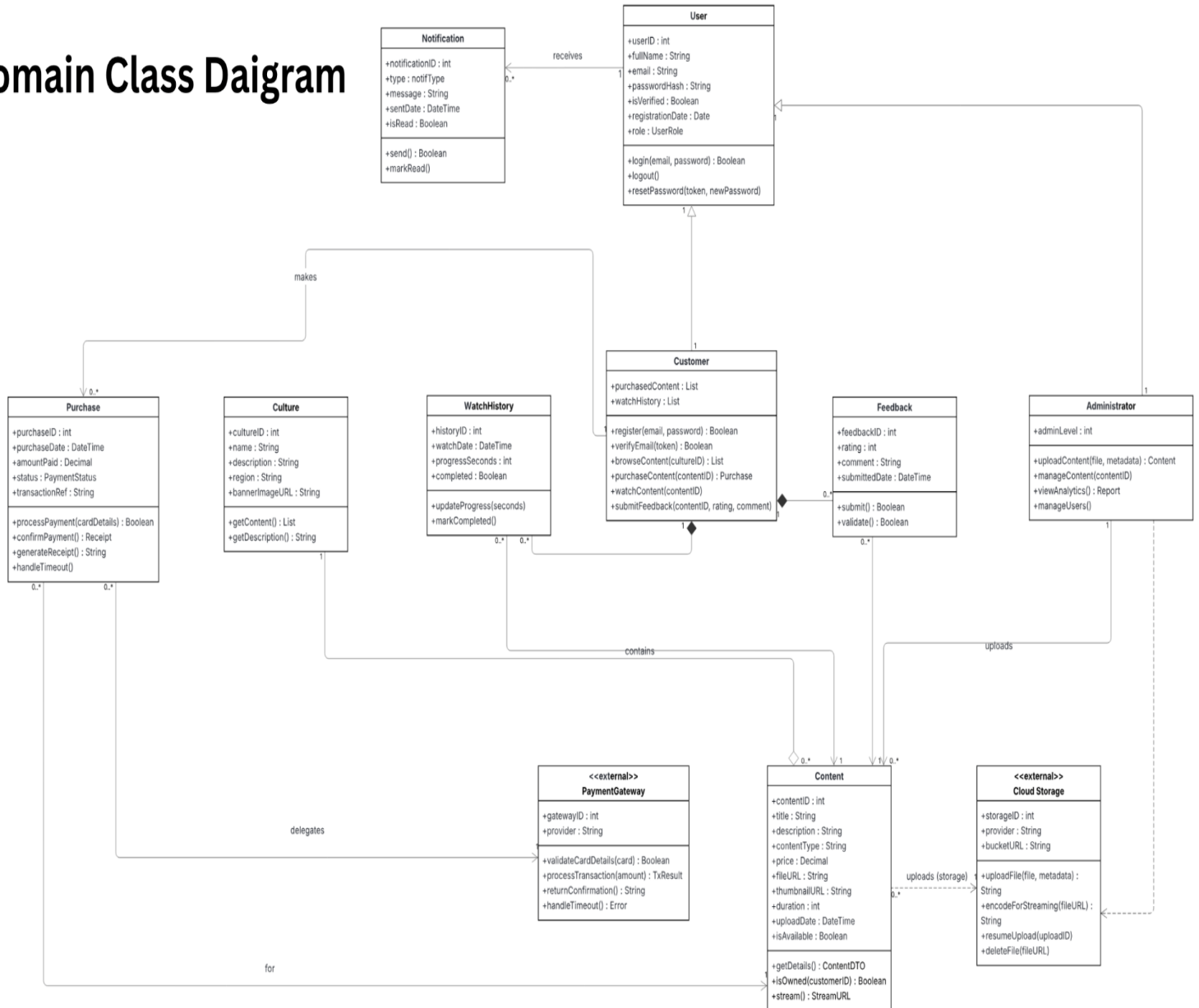


6.Domain and Data Models (Entity relationship diagram &Class domain Diagram)

Entity Relationship Diagram

Domain Class Diagram

Domain Class Diagram



7. System Feature List

Requirement	Description
Access Control	Only administrators can upload films; viewers can only browse, purchase, and watch.
Error Handling	User-friendly error messages for payment failures, upload issues, and streaming problems.
Email Notifications	Purchase receipts sent to user email; admin alerts for system issues (upload failures, payment errors).
Logging	System logs for administrator review (errors, purchases, uploads, access attempts).
Responsive Design	All interfaces function on desktop, tablet, and mobile devices.
Analytics Tracking	View counts, purchase counts, and completion rates tracked for reporting.

Requirement	Description
Data Privacy (POPIA)	Email addresses stored securely; privacy policy displayed; no unnecessary data collection.